MIL-M-542A (ORD)
28 July 1958
SUPERSEDING
JAN-M-542
18 December 1947

MILITARY SPECIFICATION

MAGNESIUM STEARATE (FOR USE IN AMMUNITION)

1. SCOPE

1.1 This specification covers one grade of magnesium stearate.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

RR-S-366	- Sieves, Standard, Testing
UU-S-48	- Sacks, Paper, Shipping
PPP-D-700	- Drums: metal, 55-gallon (for acid and corrosive liquids)
PPP-D-729	- Drums: metal, 55-gallon (for shipment of noncorrosive materials)
PPP-D-723	- Drums, fiber

MILITARY

JAN-P-112	- Packaging and Packing for Overseas Shipment, Drums Plywood

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

FSC 6810

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE PUBLICATION

Consolidated Freight Classification Ratings, Rules, and Regulations.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification Rules and Container Regulations.

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N. W., Washington 6, D. C.)

3. REQUIREMENTS

3.1 Chemical properties.
3.1.1 Moisture. - The moisture content shall be 6.5 percent, maximum (max.) when determined as specified in 4.4.1.
3.1.2 Water-soluble salts. - The water-soluble salts shall be 0.2 percent, max. when determined as specified in 4.4.2.1.
3.1.3 Magnesium. - The magnesium content shall be 4.0 ± 0.2 percent when determined as specified in 4.4.2.3.
3.1.4 Alkalinity. - The alkalinity shall be 0.03 percent, max. when determined as specified in 4.4.2.4.
3.1.5 Mineral acidity. - Magnesium stearate shall show no evidence of mineral acidity when determined as specified in 4.4.2.5.
3.1.6 Total stearate content as stearic acid. - The total stearate content as stearic acid shall be 97 percent max. and 92 percent min. when determined as specified in 4.4.2.7.
3.1.7 Iodine number (No.) of fatty acid. - The melting point of fatty acid shall be 55°C. min. when determined as specified in 4.4.2.9.
3.2 Physical properties.
3.2.1 Melting point. - The melting point shall be 130 degrees (°) Centigrade (C.), minimum (min.) when determined as specified in 4.4.2.2.
3.2.2 Grit. - Magnesium stearate shall be free from grit when tested as specified in 4.4.2.6.

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3.2.3 Melting point of fatty acid. - The melting point of fatty acid shall be 55°C. min. when determined as specified in 4.4.2.8.
3.2.4 Granulation. - Magnesium stearate shall comply with the granulation requirements shown in Table I, when tested as specified in 4.4.3.

Table I - Granulation requirements

Sieve No.	Percent passing specified sieve, min.
100	100
200	95

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot. -
4.1.1 Initial production lot. - The initial production lot shall be the first of a series of production lots as defined in 4.1.2. In addition, should any change in chemical process or physical mixing process or production equipment occur at any time during the processing phase of the magnesium stearate, the first lot produced thereafter under the altered conditions shall be considered to be anew initial production lot and shall be treated thereas.
4.1.2 Production lot. - A production lot shall consist of magnesium stearate from not more than one manufacturing batch. A batch is defined as that quantity of magnesium stearate which has been subjected to the same unit chemical process or physical mixing process intended to make the final product substantially uniform.
4.2 Sampling. -
4.2.1 Sampling for initial production lot. - The contractor shall submit one 4 ounce sample of the magnesium stearate to a Government approved laboratory for testing as specified in 4.4.
4.2.2 Sampling for production lot. - Ten percent of the containers in the lot, but in no case more than 10 nor less than 3 containers, shall be selected so as to be representative of the lot. If there are less than 3 containers in the lot, all the containers shall be sampled. The material shall be mixed to a uniform consistency throughout, and approximately 8 ounces of it shall be taken from each selected container. The primary samples shall be placed in separate airtight containers, labeled so that the container from which each sample was taken can be identified. A composite sample, of approximately 1 pound shall be made from equal portions of the primary samples. The composite sample shall be thoroughly mixed and placed in an airtight container, labeled to show the name of the material, manufacturer, plant, contract or purchase order number and lot size. All acceptance tests shall be made on the composite sample. However, if it becomes apparent during sampling that the lot is not uniform it may be required that any primary sample be tested for compliance with the requirements of the specifications. All primary samples shall be held for possible future examination should the composite sample fail to comply with the requirements.
4.3 Inspection. -

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4.3.1 General. - All tests and inspections shall be performed by the contractor except those tests and/or inspections specifically reserved for the Government as stated herein and/or by the contract.

4.3.1.1 Test equipment and inspection facilities. - The manufacturer shall furnish and maintain all test equipment and inspection facilities for making all inspection, and shall maintain complete inspection records. Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The manufacturer shall establish calibration of test equipment to the satisfaction of the Government.

4.3.1.1.1 Commercial facilities. - Manufacturers not having test equipment and facilities satisfactory to the Government may engage the services of a commercial laboratory acceptable to the Government.

4.3.2 Certification (applicable to all production lots except the initial production lot). - The contractor shall certify that the magnesium stearate complies with the applicable requirements for the following determinations:

a. moisture content
b. water soluble salts
c. melting point
d. magnesium
e. alkalinity
f. presence of mineral acidity
g. presence of grit
h. total stearate contents
i. melting point of fatty acid
j. iodine number of fatty acid
k. granulation

The certificate furnished to the Government inspector, prior to or at the time of delivery showing tests and inspections conducted and results obtained, shall be signed by a responsible agent of the certifying organization and shall be accompanied by evidence of the agent's authority to bind his principal. The Government reserves the right to check test material submitted by the contractor on certification.

4.3.3 Packing and marking. - It shall be ascertained that the packing of the magnesium stearate and marking of the containers conform to this specification.

4.4 Test procedures. -

4.4.1 Determination of moisture. - An accurately weighed portion of approximately 2 grams (gm) of the representative sample shall be transferred to a tared glass weighing dish. The dish and contents shall be heated in an oven at 100° to 105° C. for 2 hours, cooled in a desiccator, and weighed. The loss in weight shall be calculated to percent of moisture in the sample. The residue shall be retained for the determination of melting point. (see 4.4.2.2.2)

4.4.2 Properties. -

4.4.2.1 Determination of water-soluble salts. - An accurately weighed portion of approximately 1 gm. of the representative sample shall be transferred to a tared

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platinum crucible. The crucible and contents shall be heated gently until frothing ceases, then the temperature shall be raised and the contents shall be ignited until all the carbonaceous matter has been consumed. The crucible and contents shall be cooled in a desiccator and weighed, placed in 250 milliliter (ml.) beaker, covered with 100 ml. of distilled water, and boiled gently for 15 minutes. The solution shall be filtered through an ashless filter paper, and the crucible and beaker washed, decanting the washings through the filter. The filter paper shall be placed in the crucible, dried and ignited until all carbonaceous matter is consumed. The crucible and contents shall be cooled in a desiccator, and weighed. The weight after extraction with water shall be subtracted from the weight before extraction, and the difference shall be calculated to percent of watersoluble salts in the sample on a moisture-free basis.

4.4.2.2 Determination of melting point. -

4.4.2.2.1 Apparatus. - A melting point bath shall be set up equipped with a mechanical stirrer, and a source of heat than can be easily regulated. A beaker of about 1 liter capacity, about three-fourths full of clean colorless paraffin oil, may be used for the melting point bath. An accurately calibrated thermometer graduated in 0.2° C. divisions shall be suspended in the bath so that the bulb is not less than 1 1/2 inches from the bottom of the bath. If the mercury column is not completely immersed at the temperature of the observed melting point, another thermometer shall be suspended at about one-half inch from and with its bulb about the height of the middle of the exposed column of the first thermometer.

4.4.2.2.2 Determination. - A small portion of the dry sample reserved from the moisture determination shall be ground to a fine powder and transferred to a capillary melting point tube. A thin-walled capillary tube long enough to extend above the top of the bath shall be used. The tube shall have an inside diameter of about 1 millimeter (mm.) and be sealed at one end. The tube shall be filled with the dried, ground sample to a depth of about 5 mm., compacting the sample by tapping. The tube shall be fastened to the standardized thermometer so that the lower end of the tube is in contact with the bulb of the thermometer. The stirrer shall be started and the bath heated rapidly to about 15° C. below the specified melting point, then the heat source shall be adjusted so that the rise in temperature does not exceed 1° C. per minute. The temperature shall be observed and recorded at which the material in the tube first appears to become completely liquified. When a total immersion thermometer is used and part of the liquid column is not immersed, the following correction shall be added to the observed melting point:

$$\text{Correction, } ^\circ\text{C.} = N(T-t)\ X.00016$$

where:

N = No. of degrees in the exposed mercury column.
T = uncorrected melting point.
t = average temperature of the exposed column as indicated by the second thermometer.

4.4.2.3 Determination of magnesium. - An accurately weighed portion of approximately

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2 gm. of the representative sample shall be transferred to a 250 ml. beaker. The sample shall be thoroughly wet with 95 percent ethyl alcohol and any lumps which remain shall be broken up with a stirring rod. A 100 ml. portion of distilled water shall be added to the beaker, then 10 ml. of concentrated nitric acid shall be added. The mixture shall be stirred for about 2 minutes and transferred to 250 ml. separatory funnel, using a stream of cold distilled water to make certain that all solid matter is transferred to the funnel. Fifty ml. of ethyl ether shall be added to the separatory funnel. The funnel shall be stoppered and shaken vigorously for a few seconds. The funnel shall then be inverted and the pressure which has been built up in the funnel released, by carefully opening the stopcock. The stopcock shall be closed and the funnel shaken vigorously again, releasing the pressure as described herein. The funnel shall be shaken and any pressure which builds up in the funnel shall be released until all the lead stearate in the mixture appears to have been dissolved. The funnel shall be placed upright, and the two layers allowed to separate. The lower aqueous layer shall be drawn off into a 400 ml. beaker being careful not to allow any of the ether layer to run into the beaker. Ten ml. of distilled water shall be added to the funnel and the aqueous layer drawn off without shaking it. This procedure shall be repeated and 25 ml. of distilled water added. The funnel shall be stoppered and shaken vigorously releasing the pressure as directed above. The aqueous layer shall be drawn off and the washing repeated with 25 ml. of distilled water. The ether layer shall be retained for the determination of total stearate (see 4.4.2.7). All washings shall be combined with the original aqueous layer in the 400 ml. beaker. The solution shall be made slightly ammoniacal, heated to boiling, and hydrogen sulfide gas shall be bubbled through it for 1 /2 hour. The precipitate shall be allowed to settle, the solution filtered through a Whatman No. 42 filter paper, or equivalent, and the filter washed with water containing ammonia and hydrogen sulfide. The combined filtrate and washing obtained shall be boiled to remove sulfides, the solution shall be made slightly ammoniacal, 5 ml. of saturated ammonium oxalate solution shall be added and the solutions boiled for at least 1 hour. The solution shall be filtered through a Whatman No. 42 filter paper, or equivalent, and residue washed with hot water. The volume of the combined filtrate and washings obtained shall be adjusted to about 200 ml., heated to boiling and 10 ml. of 10 percent ammonium hydrogen phosphate solution added. The solution shall be cooled to room temperature. With constant stirring, 25 ml. of approximately 28 percent ammonia solution shall be added slowly. The precipitate shall be allowed to settle for at least 4 hours, the solution shall be filtered through a tared Gooch crucible and the precipitate washed with 3 percent ammonia solution. The crucible and contents shall be dried, ignited at approximately 1000°C., cooled in a desiccator, and weighed. The increase in weight of the crucible shall be calculated to percent magnesium on a moisture free basis:

$$\text{Percent of magnesium} = \frac{A \times 21.84}{W}$$

where:

A = increase in weight of crucible.
W = weight of sample (on dry basis)

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The ether layer shall be retained for the determination to total stearate content (see 4.4.2.7).

4.4.2.4 Determination of alkalinity. - An accurately weighed portion of approximately 20 gm. of the representative sample from each lot shall be transferred to a 400 ml. beaker and 40 ml. of 95 percent ethyl alcohol, made neutral to phenolphthalein by the addition of 0.1 Normal (N.) sodium hydroxide, shall be added. The mixture shall be emulsified and 200 ml. of distilled water made neutral to phenolphthalein by the addition of 0.1N sodium hydroxide shall be added. The mixture shall be stirred vigorously and filtered through a Buchner funnel containing a Whatman No. 42 filter paper, or equivalent. The beaker and filter shall be washed with neutralized distilled water prepared as described above. If the solution is colored pink or red it shall be titrated with standard 0.1N hydrochloric acid solution. The alkalinity as percent of magnesium hydroxide ($Mg(OH)_2$) in the sample shall be calculated on a moisture-free basis as follows:

$$\text{Alkalinity as } Mg(OH)_2 = \frac{2.917\ AB}{C}$$

where:

A = volume of acid solution used
B = normality of acid solution.
C = weight of moisture-free sample.

The filtrate shall be retained for the determination of mineral acidity (see 4.4.2.5)

4.4.2.5 Determination of mineral acidity. - If the portion of the filtrate as described in 4.4.2.4 was found to be alkaline to phenolphthalein, the mineral acidity shall be reported as "none". If the filtrate is colorless, methyl red indicator shall be added and the color noted to determine compliance with 3.1.5. If mineral acidity is absent the solution will be yellow. If the presence of mineral acidity is indicated by the solution turning red, the solution shall be titrated with standard 0.1N sodium hydroxide solution to the methyl red end point, and the acidity as percent sulfuric acid (H_2SO_4) in the sample shall be calculated as follows:

$$\text{Acidity as } H_2SO_4 = \frac{4.904AB}{C}$$

where:

A = volume of sodium hydroxide solution used.
B = normality of sodium hydroxide solution.
C = weight of moisture free sample.

4.4.26 Determination of grit. - A 5.0 ± 0.5 gm. portion of the representative sample from each lot shall be transferred to a 250 ml. beaker and 25 ml. of 6N (1:1) hydrochloric acid shall be added. The beaker shall be covered with a watch glass and the contents of the beaker boiled gently for 1/2 hour on a hot plate. Sufficient distilled water shall be added to make the volume approximately 200 ml. and the solution again heated to boiling. The solution shall be removed from the hot plate and the hot liquid decanted through a No. 60 U. S. Standard Sieve, complying with Specification RR-S-366 transferring quantitatively any residue to the sieve by means of a stream of water.

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The screen and residue shall be washed with sufficient ether to dissolve any stearic acid remaining on the sieve. The sieve and residue shall be dried at room temperature and any insoluble material transferred to a smooth glass slide. Any particles of grit present as indicated by the persistence of a scratching noise shall be noted when the insoluble is pressed and rubbed with a smooth steel spatula.

4.4.2.7 Determination of total stearate content. - The ether layer obtained in the sample as specified in 4.4.2.3 shall be transferred to a tared 150 ml. beaker and the contents of the separatory funnel shall be washed with three 15 ml. portions of ether. The ether extract shall be evaporated to dryness under a current of clean, dry air. The beaker and contents shall be dried at 100° to 105° C. to constant weight, cooled in a desiccator, and weighed. The gain in weight shall be calculated to percent total stearate as stearate acid in the sample on a moisture free basis. The residue shall be retained for the determination of melting point of fatty acid (see 4.4.2.8).

4.4.2.8 Determination of melting point of fatty acid. - The fatty acid obtained in the sample as specified in 4.4.2.7 shall be used. The melting point shall be determined using the capillary tube procedure specified in 4.4.2.2.

4.4.2.9 Determination of iodine number of fatty acid.

4.4.2.9.1 Reagents.

4.4.2.9.1.1 Acetic acid for Wijs solution. - Glacial acetic acid free from reducing impurities shall be used, with a melting point of 14.8±0.1° C. (containing 99 percent acetic acid). The acetic acid shall be tested and prepared as follows: If these requirements are not complied with, the results of the iodine number determination will be unreliable.

4.4.2.9.1.1.1 Test for reducing impurities (potassium permanganate test). - Two ml. of the acetic acid shall be diluted with 10 ml. of distilled water. One-tenth ml. of 0.1N potassium permanganate shall be added and the solution maintained at a temperature of 24°±3° C. The acid shall be considered satisfactory with respect to reducing impurities for use in this determination if the pink color is not completely discharged at the end of 2 hours.

4.4.2.9.1.1.2 Melting point determination. - A 6-inch test tube shall be filled two-thirds full of the acetic acid and a standardized thermometer inserted into the acid with a cork stopper that fits into the test bulb. The amount of acid shall be at least double the quantity required to cover the bulb of the thermometer when the bottom of the latter is one-half inch from the bottom of the test tube. This tube shall be suspended within a larger test tube by means of a cork. The acid shall be cooled by immersing the assembly from the ice bath, the acid stirred rather vigorously for a few minutes, thus causing the supercooled liquid to crystallize and give a mixture of liquid and acid crystals. The thermometer readings shall be taken every 15 seconds and the true melting point considered that temperature at which the reading remains constant for at least 2 minutes. In case the melting point is found to be less than 14.8° C., the acid shall not be used. Should the melting point be greater than 14.8° C., small quantities of distilled water shall be added to reduce the melting point to 14.8° C. The melting point shall be determined after each addition of water, thus making sure that the final melting point of the acid is 14.8°±0.1° C.

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4.4.2.9.1.2 Chloroform. - United States Pharmacopoeia (U. S. P.) grade, acetone-free, chloroform.

4.4.2.9.1.3 Wijs solution. - The preparation of iodine monochloride solution shall present no great difficulty, but it should be done with care and accuracy in order to obtain satisfactory results. There shall be in the solution no sensible excess, either of iodine or more particularly of chlorine over that required to form the monochloride. This condition shall be most satisfactorily attained by dissolving in the whole of the acetic acid to be used, the requisite quantity of iodine, using gently heat to assist the solution if it is found necessary. Iodine shall be dissolved in glacial acetic acid in the proportion so that 18 gm. of iodine will be present in 1,000 ml. of solution. A small portion of this solution shall be set aside while pure, and dry chlorine is passed into the remainder until the halogen content of the solution is doubled. Ordinarily it will be found that by passing the chlorine into the main part of the solution until the characteristic color of free iodine has just been discharged, there will be a slight excess of chlorine which will be corrected by the addition of the requisite amount of the unchlorinated portion until all free chlorine has been destroyed. A slight excess of iodine will do little or no harm but excess of chlorine should be avoided. As an example, if the titration of 20 ml. of the original iodine-acetic acid solution required 22 ml. of 0.1N thiosulfate, 20 ml. of the finished Wijs solution should require between 43.5 and 44 ml. of the thiosulfate solution.

4.4.2.9.1.4 Starch solution. - A mixture of 2 gm. of soluble starch, several milligrams (mg.) of mercuric iodide preservative, and a little water shall be prepared. The mixture shall be added slowly to 500 ml. of boiling water, boiled for 5 minutes, and then cooled.

4.4.2.9.1.5 Standard sodium thiosulfate solution. - Reagnet-grade sodium thiosulfate shall be dissolved in distilled water, that has been well boiled to free it from carbon dioxide, in the proportion of 24.83 gm. of crystallized sodium thiosulfate to 1,000 ml. of solution. One-tenth gm. sodium carbonate shall be added and allowed to stand overnight before standardizing. The solution shall be standardized by any suitable standard method and preserved in a stock bottle provided with a guard tube filled with soda lime.

4.4.2.9.1.6 Potassium iodide solution. - Potassium iodide free from iodate, shall be dissolved, in distilled water in the proportion of 150 gm. of potassium diodide to 1,000 ml. of the solution.

4.4.9.2 Procedure. - An accurately weighed portion of approximately 1 gm of the representative sample of the fatty acid shall be transferred to a 250-ml. bottle having a well-ground glass stopper, or an Erlenmeyer flask having a specially flanged neck for the iodine test. Ten ml. of chloroform shall be added to each of the two empty flasks like that used for the sample. By means of a pipet, 20 ml. of the Wijs solution shall be added to each flask and left to stand with occasional shakings for 60±5 minutes in a dark place at a temperature of from 21° to 23° C. At the end of this period 10 ml. of the 15 percent potassium-iodide solution and 100 ml. of water shall be added to the flask, and the solution titrated with standard sodium thiosulfate, using starch as the indicator. The titration on the two blank tests shall agree within

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0.1 ml. From the difference between the average of the blank titrations and the titration of the sample the iodine number of the material shall be calculated as follows:

$$\text{Iodine number of fatty acid} = \frac{12.7N\,(V-v)}{W}$$

where:

N = normality of sodium thiosulfate solution.
V = ml. of sodium thiosulfate solution used to titrate the blank.
v = ml. of sodium thiosulfate solution used to back titrate the sample solution.
W = gm. of the sample.

4.4.3 Determination of granulation. - A tared 3 inch No. 100 U. S. Standard sieve complying with Specification RR-S-366 shall be placed on a tared 3 inch No. 200 U. S. Standard sieve. A weighed portion of 10 gm. of the representative sample shall be transferred to a 1000 ml. beaker and wet thoroughly with distilled water containing a wetting agent. The mixture shall be poured on the top sieve, and as much of the solid material as possible shall be brushed and washed through. After thorough washing, the top sieve shall be detached from the bottom sieve and as much as possible of the material remaining shall be brushed and washed through the No. 200 sieve. The sieves shall be heated in an oven at 105° to 110° C. until dry, cooled in a desiccator, and weighed. From the weight of material held on the sieve, the percent of sample passing through each sieve shall be calculated.

4.4.4 Rejection (applicable to initial production lot only). - The initial production lot shall be rejected if the sample fails to comply with any of the requirements specified. In the event of rejection, the first subsequent lot processed thereafter shall be considered the initial production lot and subject to the specified testing procedure. This policy shall be continued until such time that an initial production lot has been found which complies with the requirements.

4.4.5 Rejection (all production lots other than initial production lot. - The lot shall be rejected if the sample fails to comply with any of the requirements specified, whether tested by the contractor or check tested by the Government.

5. PREPARATION FOR DELIVERY

5.1 Packing (see 6.2)

5.1.1 Level A. - Magnesium stearate shall be packed in 50 pound capacity 5 ply, multiwall paper shipping sacks complying with style 2X of Specification UU-S-48, or in barrels or drums conforming to Specifications PPP-D-700, PPP-D-729, PPP-D-723, and JAN-P-112.

5.1.2 Level C. - Magnesium Stearate shall be packed for shipment in conformance with Consolidated Freight Classification rules and container specifications for rail shipments, or with National Motor Freight classification rules and container specifications for truck shipments.

5.2 Marking. - Exterior containers shall be marked in accordance with Standard MIL-STD-129 and shall include the batch number.

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5.2.1 Level A. - Exterior containers for Level A shall be marked with the Ordnance Corps symbol and the SM group of the item.

6. NOTES

6.1 Intended use. - Magnesium stearate is intended for use as an ingredient in rocket propellant.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Selection of applicable levels of packing.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have furnished, formulated, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may be in any way related thereto:

Custodian:
Army-Ordnance Corps

Preparing Activity
Army-Ordnance Corps

Other Interest:
Army-C

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8879

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